Praktikum Regelungstechnik (RTP)

Versuch T2: Systemidentifikation

Versuch 2

2026-05-01

Gruppe 38
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Inhaltsverzeichnis

1. Preperations **3**
1.1. (aV) Explanation of pulse width modulation control for a drive. 3
1.2. (bV) Serial interface (RS232) and its key parameters 5
1.3. (cV) Using QEI to detect the direction of rotation 6
1.4. (dV) Review of documentation for Serial.ICE 7

2. Experiment T3: RetBot system identification, part I **8**
2.1. (e) Documentation of communication with the RetBot 8
2.2. (f) Analysis of the RetBot's drive train 9
2.3. Interpretation 9

A. Appendix Frontmatter **10**
A.1. Declarations 10

B. Appendix Code and Resources **13**
B.1. Code used for the analysis of the Preperations 13
B.2. Code used for the analysis of the Experiment 13

1 Preperations

1.1. (aV) Explanation of pulse width modulation control for a drive.

Pulse width modulation (PWM) is a technique used to control the power delivered to a load, such as a motor, by varying the width of the pulses in a pulse train. The duty cycle of the PWM signal determines the average voltage applied to the load, allowing for precise control of the motor's speed and torque. The important parameters of PWM include, the impulse width (the duration of the 'on' time), the period (the total duration of one cycle), the break time (the duration of the 'off' time), and the VCC span (the voltage level of the 'on' and 'off' states) as shown in Figure 1.1.
The duty cycles of the PWM signal can be adjusted to control the motor's behavior. See Figure 1.2 for example

- 10 % duty cycle means the motor receives power for 10 % of the time, and is off for the remaining 90
- 50 % duty cycle means the motor receives power for 50 % of the time, and is off for the remaining 50
- 90 % duty cycle means the motor receives power for 90 % of the time, and is off for the remaining 10

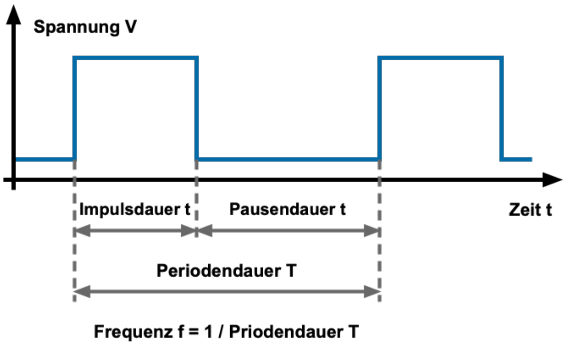

Abbildung 1.1.: Illustration of PWM parameters[1]

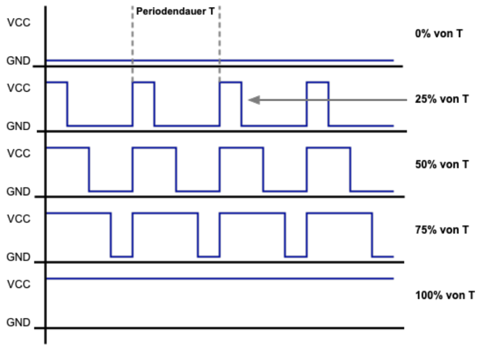
Periodendauer T
VCC
GND
0% von T
VCC
GND
25% von T
VCC
GND
50% von T
VCC
GND
75% von T
VCC
GND
100% von T

Abbildung 1.2.: Illustration of PWM duty cycles[1]

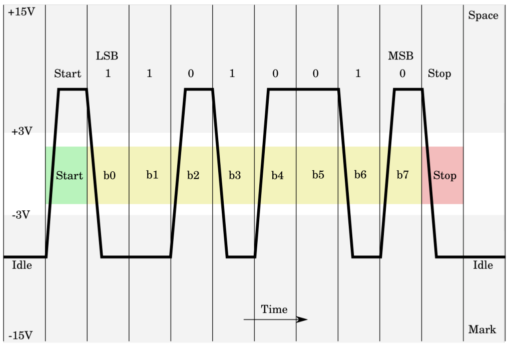

Abbildung 1.3.: By $Rs232_oscilloscope_trace.jpg$: Ktnbnderivative work: Samuel Tardieu (talk) $Rs232_oscilloscope_trace.jpg$, CC SA 1.0, https://commons.wikimedia.org/w/index.php?curid=6101703 [2]

1.2. (bV) Serial interface (RS232) and its key parameters

The RS232 interface is used for connecting devices together one sender and one receiver. Before USB it was one of the most common interfaces for connecting peripherals to computers. Both the sender and the receiver have to be configured with the same parameters for the communication to work. The key parameters are shown in figure 1.3 and include:

- Baud rate: The speed of data transmission, measured in bits per second (bps). Common baud rates include and range from 110, 9600, 19200, 38400, 57600, and 115200 bps.
- Data bits: The number of bits used to represent each character. Common settings are 7 or 8 data bits.
- Parity: A method of error checking. It can be set to none, even, odd, mark, or space.
- Stop bits: The number and configuration of stop bits used to signal the end of a data packet.
- Flow control: A method to control the flow of data between devices, which can be hardware (RTS/CTS) or software (XON/XOFF).
- Voltage levels: RS232 uses specific voltage levels for signaling, typically +3 to +15 volts for a logical '1' and -3 to -15 volts for a logical '0'.

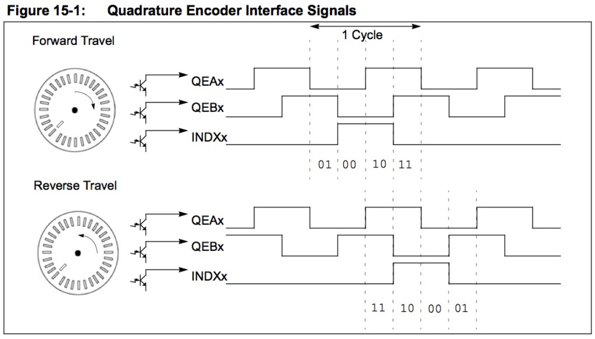

Abbildung 1.4.: Illustration of Quadrature Encoder Interface (QEI) signals[3]

1.3. (cV) Using QEI to detect the direction of rotation

Quadrature Encoder Interface (QEI) is a method used to determine the position and direction of rotation of a motor shaft. It uses two signals, typically referred to as A and B, which are generated by the encoder as the shaft rotates. [3] The Quadrature Encoder Interface (QEI) provides an interface to incremental encoders for obtaining mechanical position and speed data from rotating systems, enabling closed-loop motor control. A typical quadrature encoder consists of a slotted wheel mounted on the motor shaft and an emitter/detector module that senses the slots, producing two phase-shifted output signals, see figure 1.4, Phase A (QEAx) and Phase B (QEBx), along with an Index pulse (INDXx) that marks a reference position once per mechanical revolution. The relative phase between QEAx and QEBx encodes the direction of rotation, while a 16-bit Up/Down Counter (POSxCNT) tracks the angular position by incrementing or decrementing on each state transition. [3]
To illustrate the operating principle, consider a slotted disk mounted on the motor shaft with $N = 360$ evenly spaced slots around its circumference, sensed by two laser-based optical detectors offset by a quarter slot pitch (i.e. 90° electrical phase shift). As the disk rotates, each detector produces a square wave: detector A generates QEAx and detector B generates QEBx, with both signals running at the same frequency but offset in phase. For forward rotation, QEAx leads QEBx by 90°, so the (QEAx, QEBx) pair cycles through the state sequence $00 \rightarrow 10 \rightarrow 11 \rightarrow 01 \rightarrow 00$; for reverse rotation, the sequence is traversed in the opposite order. Each transition between states is counted, yielding $4N = 1440$ counts per mechanical revolution and an angular resolution of 0,25°. At a shaft speed of, for example, 600 rpm (10 Hz), the QEAx and QEBx signals oscillate at 3,6 kHz and the counter POSxCNT updates at 14,4 kHz, incrementing for forward rotation and decrementing for reverse. Once per revolution, the INDXx pulse asserts and can be used to reset POSxCNT, providing an absolute reference that eliminates accumulated drift.

1.4. (dV) Review of documentation for Serial.ICE

From the documentation of Serial.ICE the following key information can be extracted:

- 2 options to send data I: through the direk-in main field and II: through the 'Data to send' field.
- Data is received in the 'Main window' field and shown here.
- Logging: in the 'Serial window' click on 'LogtoFile' to log the communication to a file. The file is saved in the same directory as Serial.ICE.exe.

The necessary commands are $D25.0$, $D50.0$, $D75.0$, and $D100.0$ to set the duty cycle of the PWM signal to 25 %, 50 %, 75 %, and 100 % respectively.
Based on this objective, a standardized test plan is implemented: (i) Define communication parameters (3.0 Mbaud, 8N1, no parity/flow control), (ii) Set the drive to PWM constant mode, (iii) reset the QEI counter and document environmental conditions, (iv) for each speed level, perform a jump from a standstill and fully log the response. This provides reproducible data sets that can subsequently be evaluated in a comparable manner.

2 Experiment T3: RetBot system identification, part I

The EV3 robot was controlled via Serial.ICE using USB/virtual COM port. In constant PWM mode, symmetrical duty cycles of 25 %, 50 %, 75 %, and 100 % were initially set for both drives (d/D refer to motors 0/1). To ensure complete traceability, the following standardized procedure was followed:

- 1. Connect and configure: Plug in the EV3 via USB, open Serial.ICE, set communication parameters (e.g., 3.0 MBaud, 8N1, no parity/flow control), optionally send v (version) and r (QEI reset).
- 2. Set PWM and start logging: Select mode (m1), command desired PWM (d /D, e.g., d50.0, D50.0). Activate the log function in Serial.ICE so that commands and responses are recorded.
- 3. Positioning and test run: Disconnect the USB cable, position the robot as defined on the track. Start the test locally on the device by pressing button 2; press button 2 again to stop. (The controller block centrally controls the inputs and outputs.)
- 4. Retrieve and save measurement data: Reconnect the robot, send the command g in Serial.ICE to read the measurement data from the controller. Save/close the log file using the Log button.
- 5. Repeat for all duty cycles: Repeat steps 1 bis 4 for 25 %, 50 %, 75 %, 100 %, and, if applicable, for asymmetric settings (e.g., d25/D24, d50/D47, d75/D71, d100/D95) to improve tracking and compensate for motor differences.

2.1. (e) Documentation of communication with the RetBot

communication g - get data d25.0 - set duty cycle of motor 0 to 25
r - reset QEI counter

2.2. (f) Analysis of the RetBot's drive train

The goal of the analysis is to quantitatively describe the drive dynamics based on the recorded QEI signals (positions) and the velocities derived from them. The QEI time series were preprocessed for each duty cycle (uniform time base, offset correction), and the velocity was determined as a smoothed time derivative; this reduces high-frequency quantization and differentiation noise without significantly distorting the transient response.

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TASK: import the data into Matlab and draw one graph per drive train containing the four different conditions and

Missing figure

TASK: upload videos for two different duty-cycles moving alongside a folding rule!

2.3. Interpretation

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

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Ort, Datum und Unterschrift

Abbildungsverzeichnis

1.1. Illustration of PWM parameters[1] 3
1.2. Illustration of PWM duty cycles[1] 4
1.3. Illustration of RS232 parameters 5
1.4. Illustration of Quadrature Encoder Interface (QEI) signals[3] 6

Tabellenverzeichnis

B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations

B.2. Code used for the analysis of the Experiment

Literatur

[1] *PWM - Pulsweitenmodulation (PDM PLM PBM)*. URL: `https://www.elektronik-kompendium.de/sites/kom/0401111.htm` (besucht am 29. 04. 2026).

[2] *RS-232*. In: *Wikipedia*. 10. März 2026. URL: `https://en.wikipedia.org/w/index.php?title=RS-232&oldid=1342757218` (besucht am 29. 04. 2026).

[3] "Section 15. Quadrature Encoder Interface (QEI)". In: ().